

Benchmarking of NQCH's quantum computer

February 23, 2026

1. Report of Changes

Platform: sinq20
Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78
Calibration date: 2026-01-15 02:03:03
Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260223122252
Experiment date: 2026-02-23 12:22:52
Experiment note: temporary note!!!

Platform: sinq20
Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78
Calibration date: 2026-01-15 02:03:03
Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538
Experiment date: 2026-01-18 16:15:38
Experiment note: temporary note!!!

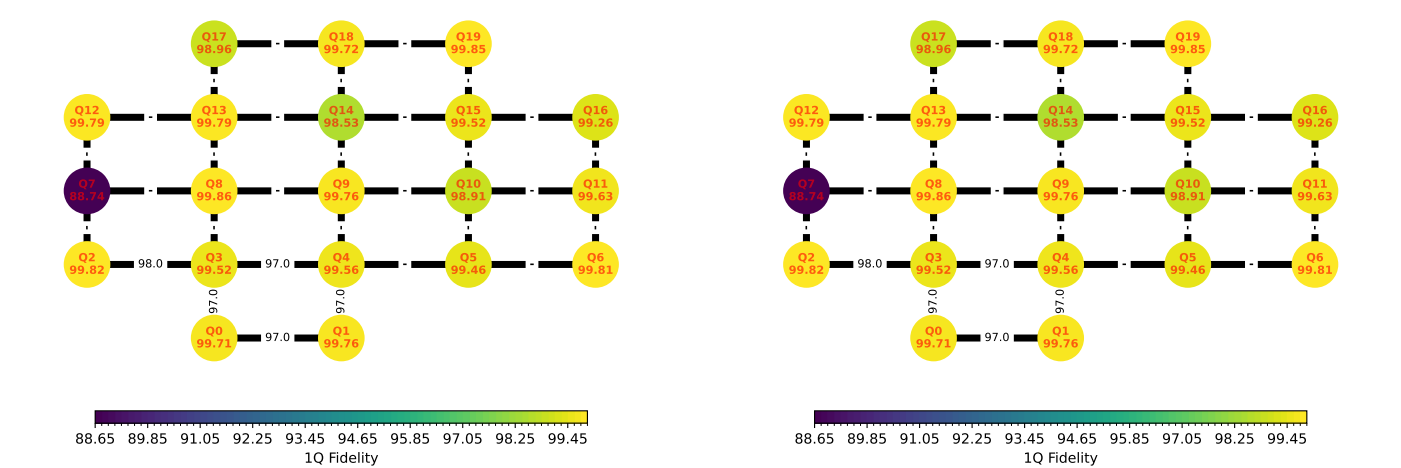
2. Version Comparison

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



88.65

89.85

91.05

92.25

93.45

94.65

95.85

97.05

98.25

99.45

1Q Fidelity

88.65

89.85

91.05

92.25

93.45

94.65

95.85

97.05

98.25

99.45

1Q Fidelity

4. Statistics

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	None	None	None	None
RO fidelity	0.794	0.777	0.777	0.927

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	None	None	None	None
RO Fidelity	0.794	0.777	0.777	0.927

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark Experiment	Qubits used	Runtime	Metric name	Metric value
Mermin	0, 2, 3	1.04 seconds	Max	1.62
Mermin	0, 2, 3	1.02 seconds	Max	-3.316
Grover - 2 qubits	2, 3	3.13 seconds	Max prob	0.882
Grover - 2 qubits	2, 3	2.36 seconds	Max prob	0.89
Grover - 3 qubits	0, 2, 3	2.97 seconds	Max prob	0.56
Grover - 3 qubits	0, 2, 3	2.43 seconds	Max prob	0.385
GHZ state preparation	0, 3, 2	2.47 seconds	Success	0.414
GHZ state preparation	0, 3, 2	3.95 seconds	Success	0.910
QFT	-	N/A	-	-
QFT	-	N/A	-	-

7. Randomized Benchmarking Results

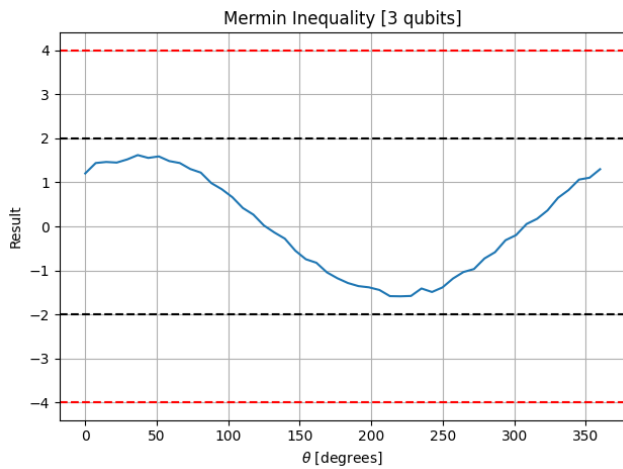
Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131
1	0.998	± 7.47e-05
2	0.998	± 7.57e-05
3	0.995	± 0.000163
4	0.996	± 0.000185
5	0.995	± 0.000147
6	0.998	± 8.34e-05
7	0.887	± 0.0287
8	0.999	± 4.48e-05
9	0.998	± 7.2e-05
10	0.989	± 0.000582
11	0.996	± 0.000218
12	0.998	± 0.000126
13	0.998	± 7.15e-05
14	0.985	± 0.00183
15	0.995	± 0.000469
16	0.993	± 0.000558
17	0.99	± 0.000502
18	0.997	± 0.00026
19	0.998	± 9.05e-05

Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131
1	0.998	± 7.47e-05
2	0.998	± 7.57e-05
3	0.995	± 0.000163
4	0.996	± 0.000185
5	0.995	± 0.000147
6	0.998	± 8.34e-05
7	0.887	± 0.0287
8	0.999	± 4.48e-05
9	0.998	± 7.2e-05
10	0.989	± 0.000582
11	0.996	± 0.000218
12	0.998	± 0.000126
13	0.998	± 7.15e-05
14	0.985	± 0.00183
15	0.995	± 0.000469
16	0.993	± 0.000558
17	0.99	± 0.000502
18	0.997	± 0.00026
19	0.998	± 9.05e-05

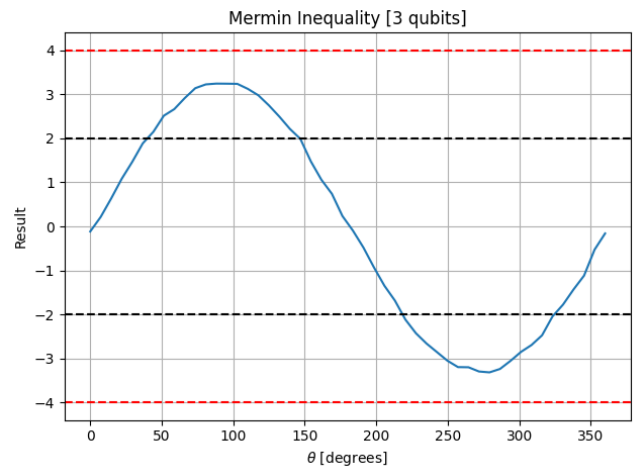
8. Mermin

Mermin's algorithm for 3 qubits.

- **Runtime:** 1.04 seconds
- **Qubits used:** 0, 2, 3
- **Max:** 1.62



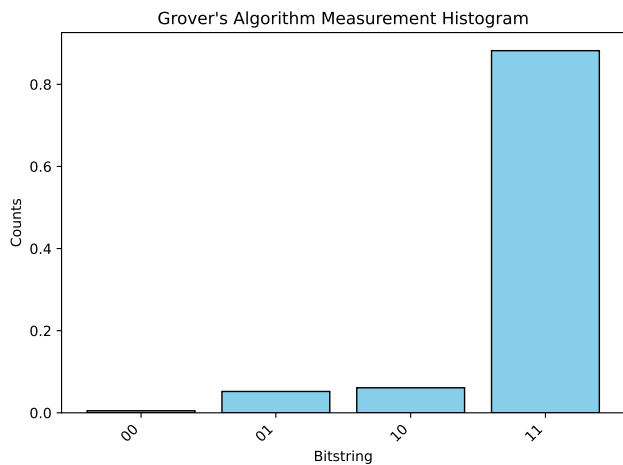
- **Runtime:** 1.02 seconds
- **Qubits used:** 0, 2, 3
- **Max:** -3.316



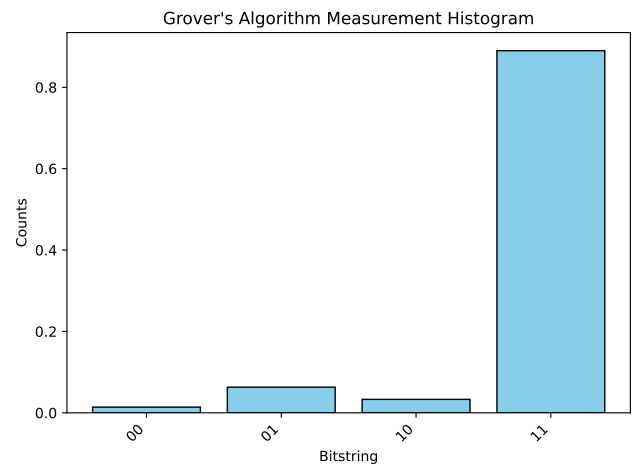
9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on `simq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in `[[2, 3]]`.

- **Runtime:** 3.13 seconds
- **Qubits used:** 2, 3
- **Max prob:** 0.882



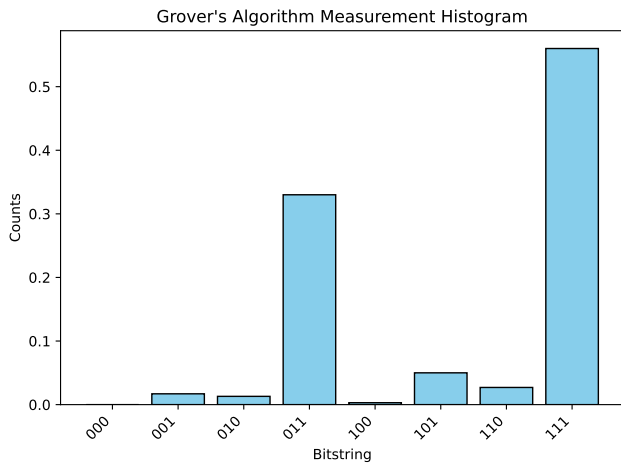
- **Runtime:** 2.36 seconds
- **Qubits used:** 2, 3
- **Max prob:** 0.89



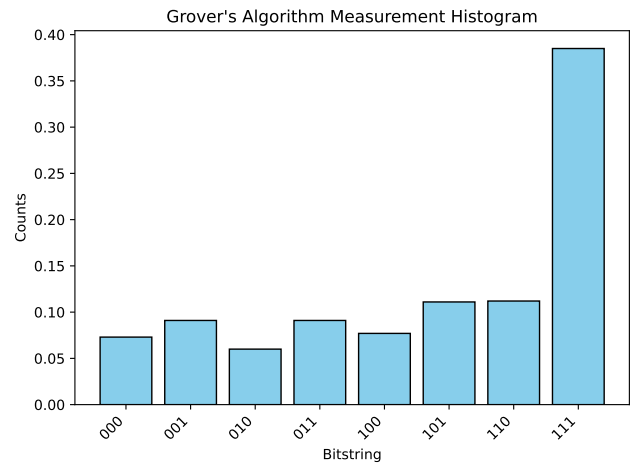
10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in $[[0, 3], [2, 3]]$.

- **Runtime:** 2.97 seconds
- **Qubits used:** 0, 2, 3
- **Max prob:** 0.56



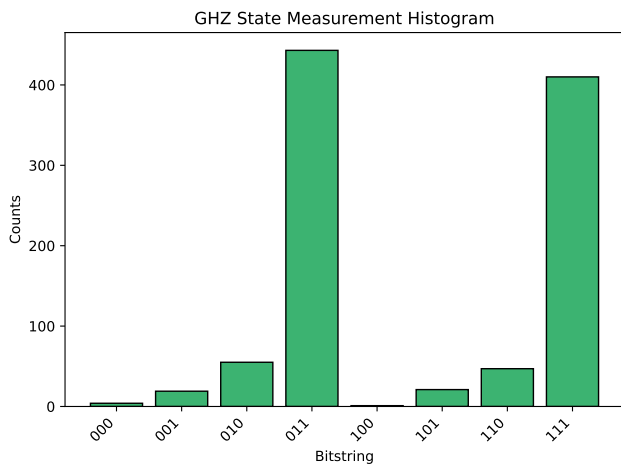
- **Runtime:** 2.43 seconds
- **Qubits used:** 0, 2, 3
- **Max prob:** 0.385



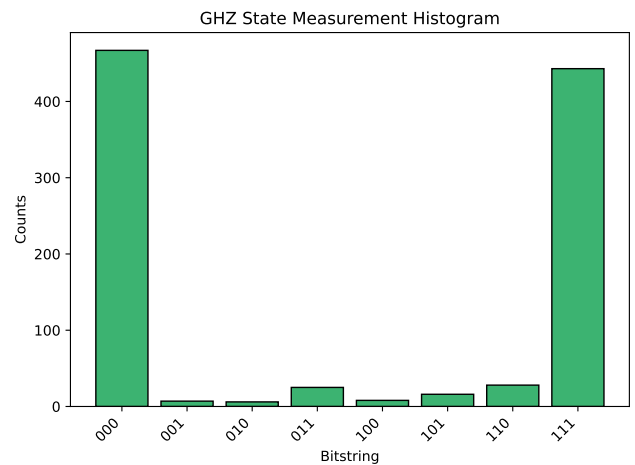
11. GHZ state preparation

GHZ circuit with 3 qubits executed on sinq20 backend with 1000 shots.

- **Runtime:** 2.47 seconds
- **Qubits used:** 0, 3, 2
- **Success:** 0.414



- **Runtime:** 3.95 seconds
- **Qubits used:** 0, 3, 2
- **Success:** 0.910



12. QFT Plots

Implementation of the Quantum Fourier Transform on three qubits with manual transpilation. The number of gates is 66, the depth of the circuit is 40

- **Runtime:** N/A
 - **Qubits used:** -
- **Runtime:** N/A
 - **Qubits used:** -

